

Frontend Development Roadmap

This roadmap will guide you through the journey of becoming a Frontend Developer by learning HTML, CSS, JavaScript, Tailwind CSS, and ReactJS. It is structured in a step-by-step manner with concepts, explanations, and examples.

HTML (HyperText Markup Language)

HTML is the backbone of every webpage. It provides the structure and semantic meaning of content. Examples include using `<h1>` for headings, `<p>` for paragraphs, `<a>` for links, and `<img>` for images. You should also learn semantic tags like `<header>`, `<nav>`, `<section>`, and `<footer>` for better structure.

- Hello World
- Visit Site

CSS (Cascading Style Sheets)

CSS is used to style and design webpages. You can control colors, fonts, spacing, and layouts. Important concepts include the Box Model, Flexbox, and Grid for responsive design.

- `h1 { color: blue; }`
- `.container { display: flex; gap: 20px; }`

JavaScript (ES6+)

JavaScript adds interactivity and dynamic behavior to webpages. You can manipulate the DOM, handle user inputs, and fetch data from APIs. Modern ES6 features include `let/const`, arrow functions, and promises.

- `document.getElementById('btn').addEventListener('click', () => alert('Clicked!'));`
- `fetch('https://api.example.com/data').then(res => res.json()).then(data => console.log(data));`

Tailwind CSS

Tailwind is a utility-first CSS framework that allows you to style elements quickly using predefined classes. It helps in building responsive designs without writing custom CSS from scratch.

- Click Me
- ...

ReactJS

React is a powerful JavaScript library for building modern user interfaces. It is based on components and hooks like `useState` and `useEffect`. React allows you to create single-page applications with dynamic content.

- `function App() { const [count, setCount] = React.useState(0); return ( setCount(count+1)}>Count: {count}); }`
- `} />`